

A Simple Tutorial on Logits

What Is a Logit?

A **logit** is the **raw score** produced by the model *before* softmax converts the scores into probabilities.

For example, suppose the possible classes are:

[World, Sports, Business, Sci/Tech]

Example: Raw Logits

The model might produce:

$$\text{logits} = [1.2, 0.5, 3.8, 0.9]$$

These are **not probabilities** — they can be negative, positive, and don't need to sum to 1.

After applying **softmax**:

$$\text{softmax}(\text{logits}) = [0.06, 0.03, 0.87, 0.04]$$

Now they *are* probabilities, and they sum to 1.

Logit = raw confidence score before softmax
